

Laboni Rahman

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LinkedIn | GitHub | Portfolio | CF

Education

Khulna University of Engineering & Technology , BSc in Computer Science & Engineering	Mar 2022 – Jul 2026
• CGPA: 3.41/4.00	
Barishal Government Women's College , HSC	Jul 2018 – Jul 2020
• GPA: 5.00/5.00	
Barishal Government Girls' High School , SSC	Jan 2008 – Jun 2018
• GPA: 5.00/5.00	

Research Interests

- Natural Language Processing (NLP) and Large Language Models (LLMs):** Text processing, sentiment analysis, machine translation, and transformer-based architectures (BERT, GPT) for tasks such as chatbots, text generation, and semantic understanding
- Machine Learning (ML) and Deep Learning (DL):** Supervised and unsupervised learning, neural networks, computer vision, model training and evaluation, and applications in prediction, classification, and data-driven decision making
- Web Development:** Full-stack development using modern frameworks and technologies (HTML, CSS, JavaScript, React, Node.js), RESTful APIs, database integration, and deployment of scalable web applications

Project Highlights

HemoCam: Smart Phone App for Non-invasive Hemoglobin Estimation <i>Java, Python, Scikit-learn</i>	GitHub	2025
- Developed a research-based mobile application for non-invasive hemoglobin estimation using skin and nail image analysis, eliminating the need for traditional blood tests - Designed the front-end using Java and implemented backend machine learning models (Scikit-learn with RobustScaler and ElasticNet) for accurate prediction - Trained on a dataset of 250 RGB images, 4800 hand images, and 15000+ annotated hand regions; integrated object detection to distinguish non-hand inputs and improve system robustness		
Hill Climbing Game: Human vs AI <i>Python, PyGame, AI Algorithms</i>	GitHub	2025
- Developed an interactive game showcasing three AI strategies—Heuristic Search, Adversarial Search, and Fuzzy Logic—enabling real-time competition between human and AI agents - Implemented decision-making algorithms including heuristic scoring for optimal path selection, adversarial prediction for opponent blocking, and fuzzy logic for dynamic risk adaptation across game phases - Designed a PyGame-based UI with real-time AI strategy visualization, move evaluation panel, and gameplay statistics; integrated dynamic obstacle generation with guaranteed solvable paths		
Voice Navigator: Voice-Controlled Web Browser Interface <i>JavaScript, Node.js, Web Speech API</i>	GitHub	2026
- Developed an accessibility-focused web navigation system enabling hands-free browser control using real-time voice commands, improving usability for diverse users - Implemented multilingual speech recognition with customizable command mapping and real-time feedback for seamless interaction and enhanced user experience - Built using modern web technologies with support for navigation commands such as browsing, scrolling, and page control; designed for extensibility and open-source collaboration		
Online Grocery Shopping Platform <i>Web Development, Full-Stack</i>	GitHub	2024
- Developed a responsive full-stack grocery shopping platform with optimized UI for seamless user experience across mobile, tablet, and desktop devices - Implemented search and browsing functionality with keyword-based queries and categorized product organization for efficient item discovery - Designed core e-commerce features including cart management, secure checkout with multiple payment options, and user profile handling for personalized data and preferences		
Online Shopping System: C++ OOP Project <i>C++17, STL, Object-Oriented Programming</i>	GitHub	2023
Developed a fully functional console-based e-commerce system implementing core OOP principles including encapsulation, inheritance, polymorphism, and abstraction		

- Engineered features such as user authentication (Admin/Customer roles), product and order management, shopping cart with real-time stock updates, and multi-method checkout system
- Designed modular class architecture using STL containers and smart pointers, with polymorphic payment processing (credit/debit card, bank transfer, cash on delivery) and dynamic order tracking

PetBuddy: Online Pet Shop Android Application | *Java, Android, Design Patterns*

GitHub 2024

- Collaboratively developed an Android-based pet shop application with scalable and maintainable architecture using software design patterns
- Implemented core features including user authentication, pet browsing, shopping cart, order placement, and emergency hotline service for animal rescue
- Applied MVC architecture along with Singleton and Factory design patterns for efficient system design; performed unit testing and UI testing to ensure reliability

Portfolio Website | *C#, PHP Web Forms, SQL Server, JavaScript, CSS*

GitHub 2024

- Developed a full-stack personal portfolio with a responsive public-facing interface and a session-protected admin panel for CRUD management of projects and visitor messages stored in SQL Server
- Implemented secure admin authentication via database-driven credential lookup, dynamic project editing with image upload, and a contact form with persistent message storage across 4 .aspx pages

Human and Car detection and count from drone images | *Python, YOLOv8, Kaggle*

GitHub 2024

- Developed an aerial object detection and human-counting pipeline using YOLOv8 fine-tuned on the VisDrone2019 dataset to detect pedestrians, people, and cars from drone imagery
- Implemented automated counting logic with real-time visualization using OpenCV, including color-coded bounding boxes, confidence labels, and dashboard overlays for human and vehicle statistics
- Optimized training on Kaggle GPU using mixed precision (amp=True), high-resolution inputs (imgsz=960), and custom VisDrone dataset configuration while handling class-index consistency issues

Skills

Languages: C, C++, Python, Java, Swift, JavaScript, C#, SQL

Databases: Firebase RTDB/Firestore, SQL Server, MySQL

AI/ML: scikit-learn, SVM, Google Gemini, OpenCV, NumPy, pandas, TensorFlow.js, YOLO, Random Forest (RF)

Web/Backend: Next.js, FastAPI, Node.js, PHP Web Forms

Mobile: Android (Java), SwiftUI/iOS, Firebase SDK

Tools & Concepts: Git, Makefile, OMNeT++, Android Studio, OOP, Design Patterns, Encryption, Network Simulation, Linux, Operating System Design

Problem Solving: Codeforces (100+ problems), LeetCode (few problems, Medium/easy), Participated in IKPC in 2022 and got 23th position.